



Dr. Yanga's Colleges Inc.
College of Computer Studies
1st Semester, SY 2025 - 2026



Cybersecurity Risk Assessment for a DYCI Frontline Office Elida Campus

Information Assurance and Security (IAS 313)

Submitted by:

ALINEA, LEMMUEL RODGE M.
AZARCON, JOHN PAUL J.
CABURNAY, IVAN T
FRANCISCO, VINZ ANGELO R.
JOSE, PHILIP JOHNSEN P.
MENDOZA, LUKE PERRY B.
MENDOZA, MARK WILSON L.
SANTOS, AISH IZHIE REIGN C.
SANTOS, FIONA LEIGH C.
SANTOS, MARK ANDREW B.

BSIT – 3A

Submitted to:

ENGR. ALBERTO C. CRUZ JR., PCpE
Instructor

I. Executive Summary

The cybersecurity assessment of the DYCI Frontline Office in Elida Campus identified several critical risks affecting the confidentiality and integrity of student information. The office relies on digital systems such as OSIRIS, the “Frontline Gate 5” network, and physical documents used for processing student credentials, request letters, and enrollment inquiries.

The review found that the most significant risks stem from weak passwords, phishing attacks, unsecured Wi-Fi, mishandling of printed documents, and unattended workstations. These vulnerabilities expose sensitive data and could disrupt frontline operations.

To address these concerns, recommended actions include implementing strong password policies, enabling multi-factor authentication (MFA), improving phishing awareness, securing Wi-Fi configurations, regularly updating antivirus software, backing up data, and enforcing workstation auto-lock policies. These measures will strengthen the office’s cybersecurity posture and help protect student information.

II. Introduction

Background: Cybersecurity plays a vital role in educational institutions, especially in departments that process large volumes of sensitive student information. The DYCI Frontline Office serves as the primary point of contact for student inquiries, document requests, and enrollment processes. Because of its public-facing nature, it handles both digital and physical forms of data that must be protected from unauthorized access, misuse, and cyber threats. Ensuring the security of these assets is necessary to maintain student trust and ensure smooth administrative operations.

Problem Statement: As the DYCI Frontline Office becomes increasingly dependent on digital platforms such as OSIRIS and internet-based communication, its current security posture remains unclear. No recent assessment has fully identified the threats, vulnerabilities, and risks affecting its systems, network, data, and personnel. Without this evaluation, the office may unknowingly face risks such as data leakage, system intrusion, phishing attacks, and operational disruptions. These weaknesses make it difficult to prioritize security actions and leave the Frontline Office open to emerging cyber threats.

Objectives:

1. Identify key assets within the department.
2. Determine potential threats and vulnerabilities.
3. Conduct qualitative/quantitative risk analysis.
4. Propose mitigation and security improvement strategies.

Scope and Limitations:

This cybersecurity assessment covers the DYCI Frontline Office in Elida Campus, specifically focusing on:

- Hardware used in operations (workstations, printers)
- Software systems (OSIRIS, email systems)
- Network resources (Frontline Gate 5 Wi-Fi)
- Data handled (student credentials, request letters, enrollment inquiries)
- Personnel with authorized access

However, the assessment does not include:

- Other DYCI administrative offices in Elida and Marciano Campus
- Non-frontline networks or servers not connected to Frontline Gate 5
- Student-owned devices
- Full penetration testing or deep forensic analysis

The scope is limited to identifying risks within the frontline operational environment.

III. Methodology

Steps:

1. **Asset Identification** – Listing hardware, software, network resources, data assets, and personnel used in the Frontline Office.
2. **Threat Identification** – Determining potential threats such as phishing, malware, unauthorized access, and data leakage.
3. **Vulnerability Assessment** – Identifying weaknesses linked to each asset and how threats may exploit them.
4. **Risk Analysis** – Evaluating likelihood and impact to categorize risks as Critical, Moderate, or Low.
5. **Mitigation Planning** – Recommending security controls and improvements for each identified risk.
6. **Reporting** – Consolidating findings into a complete cybersecurity risk assessment report.

Tools Used:

- Interviews with Frontline Office personnel
- OSIRIS system observation
- Manual inspection of frontline workstations and physical documents
- Basic risk-rating matrix (Likelihood × Impact)

IV. Asset Identification

Asset ID	Asset Name	Type	Owner	Criticality (H/M/L)	Remarks
A-01	Frontline Workstation Computer	Hardware	Frontline Office	High	Used to access OSIRIS and process student records/inquiries
A-02	Office Printer	Hardware	Frontline Office	Medium	Used for printing request forms and documents
A-03	OSIRIS System	Software	IT Department	High	Main system for student data, requests, and enrollment
A-04	Wi-Fi Network "Frontline Gate 5"	Network	IT Department	Medium	Provides network access required to connect to OSIRIS and process online transactions.
A-05	Student Credentials	Data	Frontline Office	High	Contains personal information that requires confidentiality and protection.
A-06	Request Letters	Data	Frontline Office	High	Contains official student information
A-07	Enrollment Inquiry Records	Data	Frontline Office	High	Includes student details and transactions
A-08	Frontline Personnel	People	HR Department	High	Authorized users with system access

V. Threat Identification

Threat ID	Threat Description	Threat Source	Target Asset/s	Type (Internal/External)
T-01	Phishing Emails that trick Frontline Personnel into giving login credentials	Unknown Senders	A-01, A-03, A-05	External
T-02	Malware Infection from unsafe browsing or downloading	Internet, malicious websites	A-01, A-03	External
T-03	Unauthorized Access to OSIRIS due to weak or shared passwords	Internal Personnel or outsiders	A-03, A-05	Internal/External
T-04	Wi-Fi Network Intrusion by someone trying to	Attackers within Wi-Fi range	A-04, A-03	External

	access school systems			
T-05	Data Leakage of student credentials or request letters	Human error, accidental sharing	A-05, A-06, A-07	Internal
T-06	Hardware damage or failure of computers/printers	Wear and tear, power fluctuations	A-01, A-02	Internal
T-07	Social Engineering someone pretending to be a student/Personnel to access confidential data	Walk-in individuals	A-05, A-07, A-08	External
T-08	Denial of Service network slowdown/unavailability causing OSIRIS downtime	Network overload, attackers	A-03, A-04	External
T-09	Physical Theft of printed documents or stored data	Unauthorized persons in the area	A-06, A-07	Internal/External
T-010	Email Account Compromise due to weak passwords or improper logout	Attackers	A-06, A-08	External

VI. Vulnerability Assessment

Vulnerability ID	Description	Associated Asset	Likelihood (H/M/L)	Impact (H/M/L)	Risk Level
V-01	Weak or shared passwords used to access OSIRIS or email	A-01, A-03, A-05, A-08	High	High	High
V-02	Lack of phishing awareness leading to Personnel clicking malicious links	A-01, A-03, A-05	High	High	High
V-03	No multi-factor authentication (MFA) enabled on OSIRIS	A-03, A-05	Medium	High	High
V-04	Unsecured Wi-Fi (“Frontline Gate 5”) or outdated Wi-Fi password	A-04, A-03	Medium	High	High

V-05	Outdated antivirus or no regular patch updates	A-01	Medium	Medium	Medium
V-06	Unintentional data leakage due to mishandling of printed documents	A-06, A-07	Medium	High	High
V-07	Inadequate physical security at Frontline Office	A-06, A-07	High	Medium	High
V-08	Potential malware infection from unsafe browsing	A-01, A-03	Medium	High	High
V-09	Insufficient data backup for request letters and inquiry records	A-06, A-07	Medium	High	High
V-010	Improper logout or unlocked workstation allowing unauthorized access	A-01, A-03	High	High	High

VII. Risk Analysis

Threat	Vulnerability	Likelihood	Impact	Risk Rating	Recommended Action
Phishing attack	V-02: Lack of phishing awareness	High	High	Critical	Conduct phishing awareness training; verify emails before clicking; enable email filtering
Unauthorized Access to OSIRIS	V-01: Weak/shared passwords	High	High	Critical	Implement strong password policy; enforce password changes; avoid sharing credentials
Account Compromise (No MFA)	V-03: No MFA for OSIRIS	Medium	High	Critical	Enable Multi-Factor Authentication (MFA) for all OSIRIS accounts
Wi-Fi Intrusion	V-04: Unsecured Wi-Fi / weak password	Medium	High	Critical	Update Wi-Fi password; restrict access; use WPA2/WPA3
Malware Infection	V-05: Outdated antivirus	Medium	Medium	Moderate	Update antivirus; enable auto-updates
Data Leakage (Printed Docs)	V-06: Mishandling of printed records	Medium	High	Critical	Secure documents; use trays; shred old papers

Document Theft / Viewing	V-07: Weak physical security	High	Medium	Critical	Secure documents; restrict access
Unsafe Browsing Malware	V-08: Unsafe browsing practices	Medium	High	Critical	Limit browsing; enable web filtering
Lost Records / No Backup	V-09: No data backup for letters/inquiries	Medium	High	Critical	Implement regular backups weekly or daily
Unauthorized Access to PC	V-10: Unlocked workstation	High	High	Critical	Auto-lock PCs; remind Personnel to lock screen

VIII. Risk Mitigation and Control Strategies

Risk ID	Risk Description	Proposed Mitigation	Responsible Person	Timeline	Status
R-01	Phishing attacks due to lack of awareness	Conduct phishing awareness training; enable spam filtering	IT Dept & Frontline Supervisor	2 weeks	
R-02	Weak or shared passwords in OSIRIS	Implement strong password policy; enforce regular password changes	IT Department	1 week	
R-03	No MFA causing unauthorized account access	Enable multi-factor authentication for OSIRIS and email accounts	IT Department	2 weeks	
R-04	Wi-Fi intrusion through “Frontline Gate 5” network	Update Wi-Fi password; limit users; use WPA2/WPA3 security	IT Department	1 week	
R-05	Malware infection due to outdated antivirus	Update antivirus; configure automatic updates	IT Department	3 days	
R-06	Data leakage from mishandled printed documents	Secure physical records; use document trays; shred unnecessary papers	Frontline Personnel	Immediate	
R-07	Unauthorized viewing or theft of documents	Keep sensitive documents behind counter; restrict front desk access	Frontline Office	Immediate	
R-08	Malware from unsafe browsing	Limit non-work browsing; enable web filtering	IT Department	2 weeks	
R-09	Loss of records due to no data backup	Implement weekly or daily backup of request letters & inquiry logs	Frontline Office & IT Dept	1 week	
R-010	Unauthorized access from unlocked computers	Set auto-lock (1–2 minutes); require Personnel to lock the screen	IT Dept & Frontline Personnel	Immediate	

IX. Conclusion and Recommendations

The cybersecurity assessment of the DYCI Frontline Office in Elida Campus revealed that although the office effectively supports student services such as enrollment inquiries, processing of request letters, and verification of credentials, it faces several significant cybersecurity risks. These risks stem from both digital and physical vulnerabilities, including weak password practices, susceptibility to phishing, unsecured network access, mishandling of printed documents, outdated antivirus software, and unattended workstations.

Most risks were assessed as Critical due to the high sensitivity of the student information being processed and the public-facing nature of the office. Without proper controls, these vulnerabilities may lead to data breaches, unauthorized access to OSIRIS or other systems, operational disruptions, and loss of student trust.

Despite these challenges, the assessment concluded that the majority of risks can be effectively mitigated by implementing appropriate security controls, strengthening Personnel awareness, and enforcing basic cybersecurity best practices. Improving digital and physical security measures will significantly enhance the resiliency and safety of the DYCI Frontline Office in Elida Campus.

To reduce the identified risks and enhance overall security, the following actions are strongly recommended:

1. Improve Authentication Practices

- Require strong passwords for OSIRIS and email accounts.
- Avoid sharing credentials among staff.
- Enable workstation auto-lock after short inactivity.
- (If possible) Implement two-factor authentication for sensitive systems.

2. Strengthen Network Security

- Regularly update the Frontline Gate 5 Wi-Fi password.
- Limit access only to authorized personnel and devices.
- Avoid connecting unknown devices to the network.

3. Prevent Social Engineering Attacks

- Conduct training on identifying phishing emails.
- Verify the identity of people requesting information.
- Avoid clicking suspicious links or attachments.

4. Improve Physical and Data Handling Security

- Store printed documents in secure trays or folders.
- Shred unnecessary papers containing student information.
- Keep desktops clean and avoid exposing sensitive data to walk-in students.

5. Implement Clear Security Policies

- Create guidelines for password use, system access, handling of documents, and browsing behavior.
- Conduct regular reminders and monitoring to ensure compliance.
- Encourage staff to report suspicious activities or emails.

By implementing these recommendations, the DYCI Frontline Office in Elida Campus can significantly reduce security risks and maintain reliable, secure services for students and school personnel.

X. References

Interview with DYCI Frontline Office Personnel (2025). Information about assets, processes, and current security practices.

Frontline Office Photo Documentation (2025). Visual confirmation of hardware, office layout, and workstation setup.

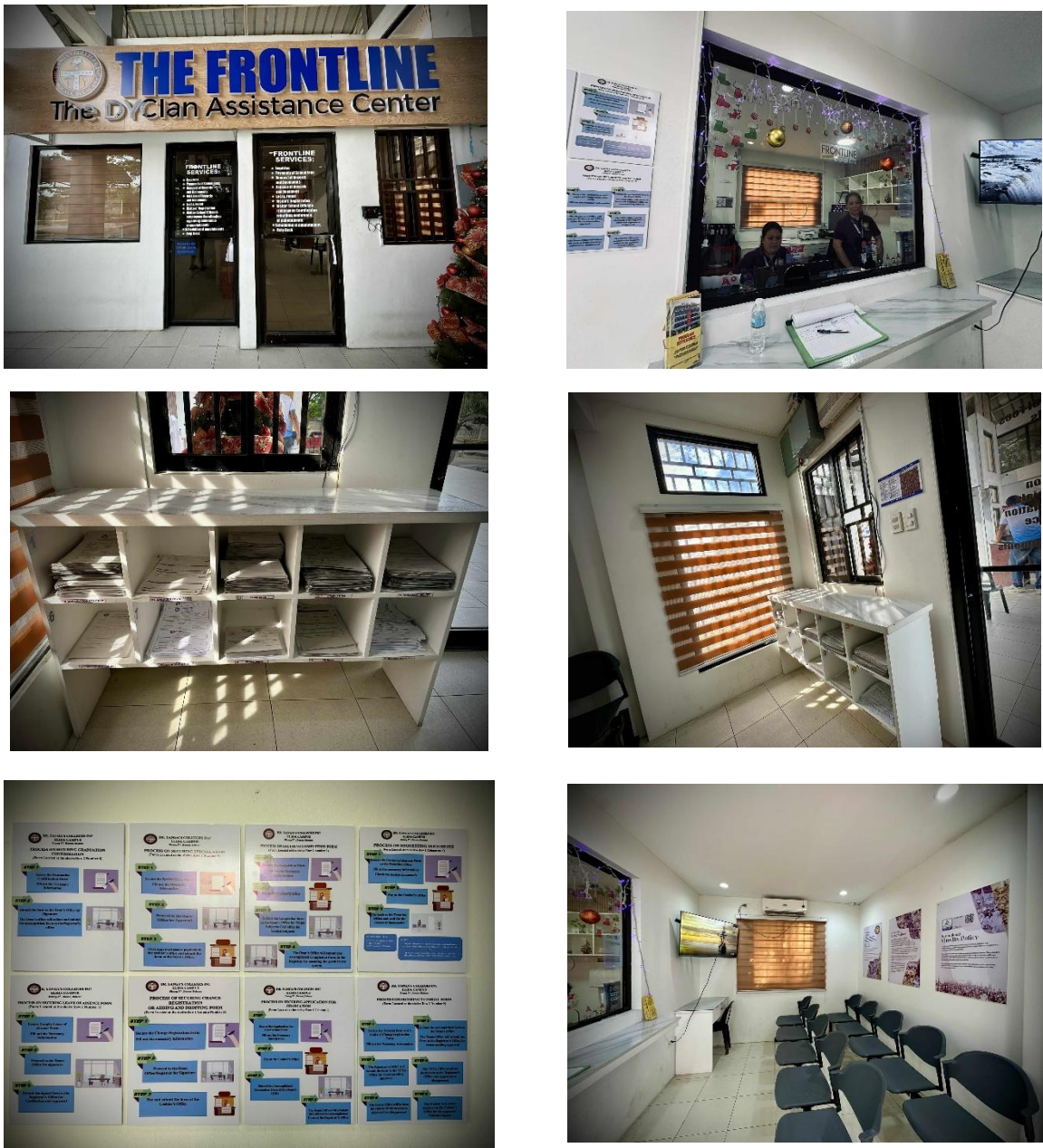
Information Assurance and Security Class Notes – Authentication Methods. (2025). Use of strong passwords and access control.

Information Assurance and Security Class Notes – Network Security. (2025). Wi-Fi security practices, secure configurations, and safe network access.

Information Assurance and Security Class Notes – Social Engineering. (2025). Phishing risks and prevention techniques.

Information Assurance and Security Class Notes – Security Policy and Mitigations. (2025). Guidelines for proper handling of data, workstation security, and mitigation strategies.

XI. Appendices



FINAL PROJECT
Information Assurance and Security (IAS 313)

CRITERIA	BEGINNING (0-4)	PROGRESSING (5-6)	ACCEPTABLE (7-8)	EXEMPLARY (9-10)	SCORE
Executive Summary	Missing or unclear summary; no direction provided.	Partial summary; lacks clarity or several key points.	Clear summary covering most key elements.	Excellent, concise summary capturing all major points and recommendations.	
Introduction, Objectives & Scope	Objectives unclear; limited context; scope undefined.	Basic introduction; objectives partially stated; scope vague.	Clear introduction, objectives, and defined scope.	Well-written intro with strong context; objectives precise; scope thorough and professional.	
Methodology	No precise method; missing steps; incomplete explanation.	Some steps are mentioned; they lack detail or structure.	Clear and logical methodology with most steps defined.	Comprehensive, well-structured methodology aligned with NIST/ISO standards.	
Asset Identification	Few assets listed; inaccurate or incomplete details.	Some assets are listed but missing key classifications or criticality.	Adequate and accurate asset table with classifications.	Comprehensive, well-organized asset inventory with thorough information and relevance.	
Threat Identification	Threats unclear or irrelevant; missing several key threats.	Some relevant threats identified; limited depth.	Clear list of threats with relevant descriptions.	Thorough, well-organized threat list with strong justification and relevance.	
Vulnerability Assessment	Missing or vague vulnerabilities; not linked to assets or threats.	Some vulnerabilities were identified, but they lack detail.	Clear vulnerability list with reasonable explanations.	Highly detailed vulnerabilities with clear relationships to assets, threats, and risks.	
Risk Analysis	No risk matrix or inaccurate analysis.	Fundamental risk analysis; inconsistent scoring.	Solid risk matrix with mostly accurate ratings.	Comprehensive, accurate risk analysis; clear prioritization and strong justification.	
Mitigation Strategies	Few or ineffective mitigation steps; not actionable.	Some strategies are listed; they lack feasibility or alignment.	Practical and relevant mitigation strategies for most risks.	Highly detailed, actionable, realistic mitigation plan aligned with best practices.	
Organization & Formatting	Disorganized; poor formatting; many errors.	Some organization has several formatting issues.	Well-organized with minimal mistakes.	Professional formatting, clear structure, highly readable.	
Completeness & Documentation	Many sections are missing; they lack references or diagrams.	Most sections provided; some missing elements.	Complete submission with needed sections and references.	Fully complete with tables, diagrams, references, and high-quality documentation.	
TOTAL SCORE					

Submitted by: (Name and Signature)

Alinea, Lemmuel Rodge M. /
Azarcon, John Paul J. /
Caburnay Ivan T. /
Francisco, Vinz Angelo R. /
Jose, Philip Johnsen P. /
Mendoza, Luke Perry B. /
Jose, Mark Wilson L. /
Santos, Aish Izhie Reign C. /
Santos, Fiona Leigh C. /
Santos, Mark Andrew B. /

Date: _____ Time: _____
Program-YearSec: BSIT-3A

Assessed by: (Prof/Instructor Name and Signature)

Engr. Alberto C. Cruz Jr., PCpE / _____

Date: _____ Time: _____